Akash S Kumbar

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Overview

I build learning systems that turn 3D perception into robotic action. My research focuses on spatial intelligence: learning representations of the world that support reconstruction, exploration, and navigation. I am interested in how robots recover meaningful structure from incomplete observations and use that understanding to guide their actions. I take these ideas through the full development cycle, from mathematical formulation and network design to training, evaluation, sensor integration, and deployment on physical robots.

Education

- **IIIT-Hyderabad** — Hyderabad, India
  MS (Research), Computer Science Engineering; CGPA: 9.33/10.0 — *2024 – Expected Dec 2026*
- **KLE Technological University** — Hubballi, India
  B.E., Electronics and Communication Engineering; CGPA: 8.42/10.0 — *2019 – Jul 2023*

Publications

* indicates equal contribution.

- **Beyond Frontiers: Scene-Anomaly Guided Autonomous Exploration** ↗ : IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.
  Best Paper Award, Learning Across Navigation Fields Workshop, IROS 2026.
  Authors: **Akash Kumbar**, Abhinav Raundhal, Madhava Krishna.
  We introduced SCAGE, which uses learned geometric anomalies in point clouds to select viewpoints for autonomous exploration. Across 10 HM3D evaluation scenes with simulated depth, it achieved **89.7% mean volumetric coverage** and **approximately 70% lower mean Chamfer distance than FrontierNet**; we also demonstrated the system on a physical robot.
- **Leveraging 2D Priors and SDF Guidance for Urban Scene Rendering** ↗ : Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
  Authors: Siddharth Tourani, Jayaram Reddy, **Akash Kumbar**, Satyajit Tourani, Nishant Goyal, Madhava Krishna, N Dinesh Reddy, Muhammad Haris Khan.
  We coupled signed distance functions with 3D Gaussian Splatting and image-derived depth and tracking priors to reconstruct dynamic urban scenes without ground-truth object-motion annotations. The LiDAR-free variant achieved **30.32 dB novel-view PSNR** on NOTR's Waymo dynamic-32 split, exceeding OmniRe's 29.35 dB.
- **ASUR3D: Arbitrary Scale Upsampling and Refinement of 3D Point Clouds Using Local Occupancy Fields** ↗ : Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2023.
  Authors: **Akash Kumbar**, Tejas Anvekar, Ramesh Ashok Tabib, Uma Mudenagudi.
  We learned local occupancy fields for arbitrary-scale point-cloud upsampling using a single trained model. At 4× upsampling on noisy PU-GAN inputs ($\tau = 0.02$), ASUR3D reduced **point-to-surface error by 8.3% relative to Grad-PU**, with additional validation on captured heritage geometry.
- **TP-NoDe: Topology-Aware Progressive Noising and Denoising of Point Clouds Towards Upsampling** ↗ : Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2023.
  Authors: **Akash Kumbar***, Tejas Anvekar*, Tulasi Amitha Vikrama, Ramesh Ashok Tabib, Uma Mudenagudi.
  We combined density-aware neighborhoods, topology-aware perturbations, and a pretrained denoiser to progressively upsample point clouds **without task-specific training**. On the PU-GAN benchmark, TP-NoDe reduced **Chamfer distance by 54% relative to PU-Net**.

Experience

- **Robotics Research Center, IIIT-Hyderabad** — Hyderabad, India
  Graduate Researcher; advisor: Prof. Madhava Krishna — *Jul 2023 – present*
  - **3D representation learning**: My research focuses on learning geometric structure from incomplete and noisy point clouds for reconstruction and autonomous exploration.
  - **Autonomous exploration**: For SCAGE, I developed geometric-anomaly prediction and viewpoint selection to steer exploration toward poorly reconstructed regions; evaluated on HM3D and deployed on a P3-DX with a RealSense D455.
  - **Dynamic scene reconstruction**: I contributed to implementation and integration across UGSDF, combining SDFs, Gaussian scene representations, and image-derived depth and tracking priors.
- **Center of Excellence in Visual Intelligence (CEVI), KLE Tech** — Hubballi, India
  Student Researcher; Research Intern (Jan – May 2023) — *Aug 2021 – May 2023*
  - **Heritage digitization**: Under Prof. Uma Mudenagudi, I worked on a **DST-funded project** supporting conservation and digital-twin generation (IHDS: Indian Heritage in Digital Space). My research upsampled and refined the sparse, noisy point clouds produced by photogrammetric reconstruction of captured images.
  - **Point-cloud upsampling**: I developed ASUR3D using local occupancy fields and co-developed TP-NoDe using topology-aware perturbations and a pretrained denoiser, producing **two first-author ICCV 2023 Workshop papers**

Projects

Titles link to project pages with more details.

- **Multi-Camera–LiDAR Calibration for an Autonomous Research Vehicle** ↗ : Developed intrinsic and extrinsic calibration for Bodhyaan, I-Hub Data's autonomous research vehicle at IIIT-Hyderabad, with **six FLIR Blackfly S cameras and an Ouster OS1-64 LiDAR**, achieving **camera-intrinsic RMS reprojection error below 0.05 px**. Built a hybrid ChArUco–retroreflective target visible to both sensors; extracted it from LiDAR scans using reflectivity filtering, PCA plane fitting, and DBSCAN; and solved the extrinsics across frames with EPnP–RANSAC initialization and soft-L1 robust refinement, outlier rejection, and per-frame residual diagnostics. Checked LiDAR-to-image alignment qualitatively on a campus drive.
  Tech Stack: Python, OpenCV, NumPy, SciPy, scikit-learn, Rerun
- **Dynamic-Obstacle Detection and Avoidance** ↗ : Co-developed and deployed a ROS navigation stack on a Husky A200 with a RealSense D455 and Livox Mid-70, integrating RGB-D/LiDAR detection and tracking of moving obstacles, Fast-LIO localization against a prebuilt map, global planning, and MPC-based reactive avoidance, all running onboard. The system avoided collisions in **95% of dynamic-obstacle encounters across more than 20 indoor and outdoor trials**.
  Tech Stack: ROS Noetic, Fast-LIO, RGB-D/LiDAR perception, RRT*, MPC, RViz
- **Learning-Based Point-Cloud Hole Filling** ↗ : Developed a model to detect and fill gaps in point clouds caused by occlusion and reflections. Designed a synthetic-hole generation pipeline to create training and evaluation examples.
  Tech Stack: Python, PyTorch, PyTorch3D, Open3D
- **Camera-Based Road-Hazard Detection (Bosch PRIXEL)** ↗ : Developed a real-time, camera-based object-detection system for potholes and marked or unmarked speed breakers. Collected and annotated a dataset of Indian road scenes and used it to train and evaluate the detection pipeline.
  Tech Stack: Python, PyTorch, OpenCV, ONNX
- **3D Point-Cloud Instance Segmentation** ↗ : Designed a U-Net-based architecture with a custom encoder for learning global point-cloud features. On S3DIS, achieved **67.6% semantic mIoU and 53.8% instance mAP@50 on Area 5**, and **57.7% instance mAP@50 under six-fold cross-validation**.
  Tech Stack: Python, PyTorch, PyTorch3D, Open3D
- **MAST3R-SLAM** ↗ : Extended an existing MASt3R implementation with camera-pose and point-cloud extraction utilities; inspected reconstructed geometry on KITTI sequences.
  Tech Stack: Python, PyTorch, CUDA
- **ARROU: Adversarially Robust Rank-One Unlearning for LLMs** ↗ : Co-developed an attacker-in-the-loop framework combining latent adversarial training with localized rank-one model editing for targeted unlearning in large language models. Evaluated forgetting and retained utility on Llama 3.2 1B using WMDP and TOFU benchmarks.
  Tech Stack: Python, PyTorch, Hugging Face Transformers, CUDA

Technical Skills

- **Languages**: Python, C/C++, CUDA, Embedded C
- **ML Frameworks**: PyTorch, Hugging Face Transformers, ONNX, scikit-learn
- **3D Vision & Perception**: Point-cloud processing, 3D reconstruction, neural implicit representations (SDFs), 3D Gaussian Splatting, semantic/instance segmentation, object detection and tracking
- **Robotics & Sensors**: ROS, SLAM/localization, motion planning, MPC, multi-camera and camera–LiDAR calibration, RGB-D/LiDAR fusion, real-robot deployment (Husky A200, Pioneer P3-DX)
- **Data**: Dataset collection and annotation, synthetic training-data generation
- **Tools & Simulation**: OpenCV, Open3D, PyTorch3D, NumPy, SciPy, Rerun, RViz, Gazebo, CARLA, Habitat

Teaching and Service

- **Teaching Assistant**: Mobile Robotics (CS7.503) ↗ , IIIT-Hyderabad — Monsoon 2026
- **Instructor**: RRC Summer School, IIIT-Hyderabad (2025 & 2026) — Taught deep learning for 3D vision, neural representations, and 3D Gaussian Splatting; 2026 lecture recording ↗
- **Reviewer**: ICVGIP 2026 ↗ ; SeMaNa Workshop (IROS 2026) ↗

Honors and Awards

- **Best Paper Award**, Learning Across Navigation Fields: Challenges and Future Work ↗ , IROS 2026 Workshop (Sep 2026) — Awarded for *Beyond Frontiers: Scene-Anomaly Guided Autonomous Exploration.*
- **Certificate of Appreciation**, Bosch's PRIXEL Program, Bosch Global Software Technologies — Awarded for developing a robust, real-time computer vision-based speed breaker and pothole detection model.